

Fleet investment committee

Five-year keep, replace or lease decision

Synthetic portfolio analysis. Recommendations are proposed; no employer outcomes are claimed.

Decision requested

Prefer replacement in the base case, subject to comparable vendor quotes, financing/liquidity review and tax validation. Reopen the decision under the fuel-shock case, where leasing has the lowest modeled present-value cost.

Financial case

Six replacements need \$660,000 net upfront cash after old-vehicle sale proceeds, within the illustrative \$700,000 capital limit. Base replacement has \$373,698 incremental NPV versus keeping the old fleet. Its advantage over leasing is only \$15,208 of PV cost, so small quote or residual changes may reverse the decision.

Case	Keep PV cost	Replace PV cost	Lease PV cost
Base	\$2,257,158	\$1,883,460	\$1,898,668
Fuel shock	\$2,517,974	\$2,060,006	\$2,031,806
Low utilization	\$1,835,122	\$1,531,764	\$1,546,972

Tradeoffs and alternatives

Replacement consumes upfront cash and bears residual-value risk. Leasing preserves liquidity but adds contractual payments, usage limits and vendor dependency. Keeping the fleet avoids purchase cash but exposes higher modeled maintenance and downtime.

Approval and execution

Conditions before approval

Confirm identical service capacity, included lease maintenance, vehicle availability, actual residual quotes and cash budget. Evaluate after-tax effects and lease accounting separately. Native sensitivity tests the real model discount rate and old-fleet maintenance input.

Proposed owner	Timing	Required action
Fleet + Procurement	Day 10	Obtain comparable purchase and lease quotes
Finance + Treasury	Day 15	Approve capital and liquidity conditions
Finance	Before commitment	Validate tax and residual-value assumptions
Fleet + Finance	Days 30 / 90 / 180	Review mpg, maintenance and downtime against plan

Post-decision review

The following observations are a separate synthetic review exercise. They demonstrate follow-through without claiming realized results.

Measure	Plan	Simulated observation	Response
Annual maintenance	\$54,000	\$58,000	Review repair mix
Fuel economy	7.2 mpg	6.9 mpg	Review duty cycle
Annual fleet downtime	24 days	30 days	Review availability
Net upfront cash	\$660,000	\$672,000	Reconcile invoices

Model basis and review

Calculation method

Five-year, pre-tax incremental cash-flow comparison. Year 0 includes net purchase/disposal cash or lease deposit/disposal proceeds. Operating cash costs cover maintenance, fuel and downtime with 3% annual inflation. Terminal receipts and deposit return occur at the end of year five. NPV uses end-year discounting.

Material limitations

Fuel-shock case changes fuel price, maintenance and discount rate together; it is not a fuel-only attribution. Downtime cost is assumed cash exposure, not independently observed loss. No depreciation tax shields, financing, lease-accounting entries, mileage penalties, insurance or transition costs are included.

Scenario capture and refresh

The comparison uses three sequential recalculations of one active Excel build, captured October 3, 2026. Change Assumptions D5 to select a case. Recalculate in Excel and refresh comparisons after changing inputs; PDF figures are a dated snapshot.

Ownership represented

Quincy Jones designed the analysis, built the model and controls, evaluated alternatives, and authored the decision and monitoring workflow. Proposed corporate owners are roles inside the case, not positions held by Quincy.

Forensic review

Independent calculations, reconciliation checks and input-change tests were completed before publication. The downloadable audit explains scope and engine limitations. Artifact-tool recalculation and exported caches were checked; desktop Excel was not available for an application-level test.

Sources

All operating and quote inputs are synthetic. Method reference: ACCA investment appraisal, <https://www.accaglobal.com/uk/en/student/exam-support-resources/prodipsust-study-resources/technical-articles/investment-appraisal.html>